Solution of triangle: Questions and Full Solutions

Question 01

English

Let ABC be a triangle with $\angle B - \angle C = \dfrac{\pi}{2}$, and let a, b, c denote the lengths of the sides BC, CA, AB, respectively. Then which of the following relations is TRUE?

(A) $(b^2 + c^2)^2 = a^2(b^2 - c^2)$
(B) $b^2 - c^2 = a^2$
(C) $(b^2 - c^2)^2 = a^2(b^2 + c^2)$
(D) $b^2 + c^2 = 2a^2$

Only one option is correct.

हिन्दी

माना ABC एक त्रिभुज है जिसमें $\angle B - \angle C = \dfrac{\pi}{2}$ है, तथा a, b, c क्रमशः भुजाओं BC, CA, AB की लम्बाइयाँ हैं। तब निम्न में से कौन-सा सम्बन्ध सत्य है?

(A) $(b^2 + c^2)^2 = a^2(b^2 - c^2)$
(B) $b^2 - c^2 = a^2$
(C) $(b^2 - c^2)^2 = a^2(b^2 + c^2)$
(D) $b^2 + c^2 = 2a^2$

केवल एक विकल्प सही है।

Solution

Let $C = \theta$. Since $B - C = \dfrac{\pi}{2}$, $B = \dfrac{\pi}{2} + \theta$, and $A = \pi - B - C = \dfrac{\pi}{2} - 2\theta$. For $A > 0$ we need $0 < \theta < \dfrac{\pi}{4}$; θ is the parameter to be eliminated.

By the sine law, $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = k$ (where $k = 2R$).

$\sin A = \sin\left(\dfrac{\pi}{2} - 2\theta\right) = \cos 2\theta$, $\sin B = \sin\left(\dfrac{\pi}{2} + \theta\right) = \cos\theta$, $\sin C = \sin\theta$.

Hence $a = k\cos 2\theta$, $b = k\cos\theta$, $c = k\sin\theta$.

Then $b^2 + c^2 = k^2(\cos^2\theta + \sin^2\theta) = k^2$ and $b^2 - c^2 = k^2(\cos^2\theta - \sin^2\theta) = k^2\cos 2\theta = k \cdot a$.

From the second relation $k = \dfrac{b^2 - c^2}{a}$; substituting in the first: $\dfrac{(b^2 - c^2)^2}{a^2} = b^2 + c^2$, i.e. $(b^2 - c^2)^2 = a^2(b^2 + c^2)$. So (C) is true.

Check of the others with $\theta = \dfrac{\pi}{12}$: $b^2 + c^2 = k^2$, $b^2 - c^2 = k^2\cos\dfrac{\pi}{6} = \dfrac{\sqrt{3}}{2}k^2$, $a^2 = k^2\cos^2\dfrac{\pi}{6} = \dfrac{3}{4}k^2$.

(A): $k^4 \neq \dfrac{3}{4}k^2 \cdot \dfrac{\sqrt{3}}{2}k^2$; (B): $\dfrac{\sqrt{3}}{2}k^2 \neq \dfrac{3}{4}k^2$; (D): $k^2 \neq \dfrac{3}{2}k^2$. So (A), (B), (D) are wrong.

Correct Option: (C)

Question 02

English

Let A, B, C, D be points in a plane such that C and D lie on opposite sides of the line AB, $AB = 21$, $BC = 13$, $CA = 20$, $AD = 10$ and $\angle CAD = \frac{\pi}{2}$. Then $\tan\angle CBD$ is equal to

(A) $-\frac{220}{21}$
(B) $\frac{220}{21}$
(C) $-\frac{21}{220}$
(D) $\frac{21}{220}$

Only one option is correct.

हिन्दी

माना एक तल में बिन्दु A, B, C, D इस प्रकार हैं कि C तथा D रेखा AB के विपरीत पक्षों में स्थित हैं, $AB = 21$, $BC = 13$, $CA = 20$, $AD = 10$ तथा $\angle CAD = \frac{\pi}{2}$ है। तब $\tan\angle CBD$ बराबर है

(A) $-\frac{220}{21}$
(B) $\frac{220}{21}$
(C) $-\frac{21}{220}$
(D) $\frac{21}{220}$

केवल एक विकल्प सही है।

Solution

In triangle ABC, by the cosine rule, $\cos\angle CAB = \frac{AB^2 + AC^2 - BC^2}{2 \cdot AB \cdot AC} = \frac{441 + 400 - 169}{840} = \frac{672}{840} = \frac{4}{5}$, so $\sin\angle CAB = \frac{3}{5}$.

Since C and D lie on opposite sides of AB and $\angle CAD = \frac{\pi}{2}$, the ray AB lies inside $\angle CAD$ (as $\angle CAB < \frac{\pi}{2}$), so $\angle BAD = \frac{\pi}{2} - \angle CAB$.

Compound angle: $\cos\angle BAD = \cos\left(\frac{\pi}{2} - \angle CAB\right) = \sin\angle CAB = \frac{3}{5}$.

Cosine rule in triangle ABD: $BD^2 = 21^2 + 10^2 - 2 \cdot 21 \cdot 10 \cdot \frac{3}{5} = 541 - 252 = 289$, so $BD = 17$.

Cosine rule at B: $\cos\angle ABC = \frac{441 + 169 - 400}{2 \cdot 21 \cdot 13} = \frac{210}{546} = \frac{5}{13}$, so $\sin\angle ABC = \frac{12}{13}$, $\tan\angle ABC = \frac{12}{5}$.

$\cos\angle ABD = \frac{441 + 289 - 100}{2 \cdot 21 \cdot 17} = \frac{630}{714} = \frac{15}{17}$, so $\sin\angle ABD = \frac{8}{17}$, $\tan\angle ABD = \frac{8}{15}$.

As C and D are on opposite sides of AB, $\angle CBD = \angle ABC + \angle ABD$ (their sum is less than π).

$\tan\angle CBD = \frac{\tan\angle ABC + \tan\angle ABD}{1 - \tan\angle ABC \tan\angle ABD} = \frac{\frac{36+8}{15}}{1 - \frac{96}{75}} = \frac{44}{15} \cdot \left(-\frac{75}{21}\right) = -\frac{220}{21}$.

The negative value means $\angle CBD$ is obtuse; check: $CD^2 = 20^2 + 10^2 = 500$ and $13^2 + 17^2 - 2 \cdot 13 \cdot 17\cos\angle CBD = 458 + 42 = 500$ with $\cos\angle CBD = -\frac{21}{221}$.

Correct Option: (A)

Question 03

English

In a triangle ABC, let a, b, c denote the lengths of the sides opposite to the angles A, B, C, respectively. Suppose $\dfrac{b-c}{b+c} = \dfrac{1}{3}$ and $\tan\dfrac{B}{2}\tan\dfrac{C}{2} = \dfrac{1}{3}$. Then $\tan\dfrac{B-C}{2}$ is equal to

(A) $\dfrac{\sqrt{15}}{9}$

(B) $\dfrac{\sqrt{15}}{5}$

(C) $\dfrac{\sqrt{15}}{3}$

(D) $\dfrac{3\sqrt{15}}{5}$

Only one option is correct.

हिन्दी

एक त्रिभुज ABC में माना a, b, c क्रमशः कोणों A, B, C की सम्मुख भुजाओं की लम्बाइयाँ हैं। माना $\dfrac{b-c}{b+c} = \dfrac{1}{3}$ तथा $\tan\dfrac{B}{2}\tan\dfrac{C}{2} = \dfrac{1}{3}$ है। तब $\tan\dfrac{B-C}{2}$ बराबर है

(A) $\dfrac{\sqrt{15}}{9}$

(B) $\dfrac{\sqrt{15}}{5}$

(C) $\dfrac{\sqrt{15}}{3}$

(D) $\dfrac{3\sqrt{15}}{5}$

केवल एक विकल्प सही है।

Solution

Write $x = \tan\dfrac{B}{2}$, $y = \tan\dfrac{C}{2}$, $z = \tan\dfrac{A}{2}$; all are positive since each half-angle lies in $\left(0, \dfrac{\pi}{2}\right)$.

Conditional identity: $A+B+C = \pi$ gives $\dfrac{B+C}{2} = \dfrac{\pi}{2} - \dfrac{A}{2}$, so $\tan\dfrac{B+C}{2} = \cot\dfrac{A}{2}$, i.e. $\dfrac{x+y}{1-xy} = \dfrac{1}{z}$, equivalently $xy + yz + zx = 1$.

With $xy = \dfrac{1}{3}$: $z(x+y) = \dfrac{2}{3}$, so $x + y = \dfrac{2}{3z}$.

Tangent rule (Napier's analogy): $\tan\dfrac{B-C}{2} = \dfrac{b-c}{b+c}\cot\dfrac{A}{2} = \dfrac{1}{3z}$.

Also $\tan\dfrac{B-C}{2} = \dfrac{x-y}{1+xy} = \dfrac{3(x-y)}{4}$. Hence $x - y = \dfrac{4}{9z}$ (positive, as $b > c$ gives $B > C$).

Solving: $x = \dfrac{1}{2}\left(\dfrac{2}{3z} + \dfrac{4}{9z}\right) = \dfrac{5}{9z}$, $y = \dfrac{1}{2}\left(\dfrac{2}{3z} - \dfrac{4}{9z}\right) = \dfrac{1}{9z}$.

Then $xy = \dfrac{5}{81z^2} = \dfrac{1}{3}$, so $z^2 = \dfrac{15}{81}$ and $z = \dfrac{\sqrt{15}}{9}$.

Therefore $\tan\dfrac{B-C}{2} = \dfrac{1}{3z} = \dfrac{9}{3\sqrt{15}} = \dfrac{3}{\sqrt{15}} = \dfrac{\sqrt{15}}{5}$.

Check: the data describe the triangle with $a : b : c = 3 : 4 : 2$, for which $\tan\dfrac{B}{2}\tan\dfrac{C}{2} = \dfrac{s-a}{s} = \dfrac{1}{3}$

and $\dfrac{b-c}{b+c}=\dfrac{1}{3}$.

Correct Option: (B)

Question 04

English

Let the position vectors of the points A, B, C be $\vec{a}=3\hat{i}+\hat{j}+2\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+2\hat{k}$ and $\vec{c}=5\hat{i}+\hat{j}+6\hat{k}$, respectively, and let D be the point with position vector $\dfrac{\vec{b}+2\vec{c}}{3}$. Then $\angle ADC$ is equal to

(A) $\dfrac{\pi}{4}$

(B) $\dfrac{\pi}{3}$

(C) $\dfrac{2\pi}{3}$

(D) $\dfrac{3\pi}{4}$

Only one option is correct.

हिन्दी

माना बिन्दुओं A, B, C के स्थिति सदिश क्रमशः $\vec{a}=3\hat{i}+\hat{j}+2\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+2\hat{k}$ तथा $\vec{c}=5\hat{i}+\hat{j}+6\hat{k}$ हैं, तथा D वह बिन्दु है जिसका स्थिति सदिश $\dfrac{\vec{b}+2\vec{c}}{3}$ है। तब $\angle ADC$ बराबर है

(A) $\dfrac{\pi}{4}$

(B) $\dfrac{\pi}{3}$

(C) $\dfrac{2\pi}{3}$

(D) $\dfrac{3\pi}{4}$

केवल एक विकल्प सही है।

Solution

Since $\dfrac{\vec{b}+2\vec{c}}{3}=\dfrac{1\cdot\vec{b}+2\cdot\vec{c}}{1+2}$, by the section formula D divides BC internally in the ratio $BD:DC=2:1$.

Angle B: $\overrightarrow{BA}=\vec{a}-\vec{b}=2\hat{i}+2\hat{j}$, $\overrightarrow{BC}=\vec{c}-\vec{b}=4\hat{i}+2\hat{j}+4\hat{k}$, $|\overrightarrow{BA}|=2\sqrt{2}$, $|\overrightarrow{BC}|=6$.

$\cos B=\dfrac{8+4+0}{2\sqrt{2}\cdot 6}=\dfrac{1}{\sqrt{2}}$, so $B=\dfrac{\pi}{4}$ and $\cot B=1$.

Angle C: $\overrightarrow{CA}=-2\hat{i}+0\hat{j}-4\hat{k}$, $\overrightarrow{CB}=-4\hat{i}-2\hat{j}-4\hat{k}$, $|\overrightarrow{CA}|=2\sqrt{5}$.

$\cos C=\dfrac{8+0+16}{2\sqrt{5}\cdot 6}=\dfrac{2}{\sqrt{5}}$, $\sin C=\dfrac{1}{\sqrt{5}}$, so $\cot C=2$.

m-n theorem: if D divides BC with $BD:DC=m:n$ and $\theta=\angle ADC$, then $(m+n)\cot\theta=n\cot B-m\cot C$.

With $m=2,\ n=1$: $3\cot\theta=1\cdot 1-2\cdot 2=-3$, so $\cot\theta=-1$.

As $\theta\in(0,\pi)$, $\theta=\dfrac{3\pi}{4}$.

Check: $\overrightarrow{DA}=\dfrac{-2\hat{i}+2\hat{j}-8\hat{k}}{3}$, $\overrightarrow{DC}=\dfrac{4\hat{i}+2\hat{j}+4\hat{k}}{3}$, $\cos\theta=\dfrac{-8+4-32}{\sqrt{72}\cdot 6}=-\dfrac{1}{\sqrt{2}}$.

Correct Option: (D)

Question 05

English

In a triangle ABC, $\sin A : \sin B : \sin C = 7 : 8 : 9$. Let I be the incentre of the triangle. If $CI = 2\sqrt{14}$, then the perimeter of the triangle ABC is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक त्रिभुज ABC में $\sin A : \sin B : \sin C = 7 : 8 : 9$ है। माना I त्रिभुज का अन्तःकेन्द्र है। यदि $CI = 2\sqrt{14}$ है, तो त्रिभुज ABC का परिमाप _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the sine law $a : b : c = \sin A : \sin B : \sin C = 7 : 8 : 9$. Write $a = 7k$, $b = 8k$, $c = 9k$ with $k > 0$.

Then $s = \dfrac{a+b+c}{2} = 12k$, $s - a = 5k$, $s - b = 4k$, $s - c = 3k$.

Cosine rule: $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab} = \dfrac{49 + 64 - 81}{112} = \dfrac{32}{112} = \dfrac{2}{7}$.

Half-angle formula: $\cos\dfrac{C}{2} = \sqrt{\dfrac{s(s-c)}{ab}} = \sqrt{\dfrac{12k \cdot 3k}{56k^2}} = \sqrt{\dfrac{9}{14}} = \dfrac{3}{\sqrt{14}}$; the same value follows from $\cos^2\dfrac{C}{2} = \dfrac{1+\cos C}{2} = \dfrac{9}{14}$.

Let the incircle touch CA at E. The tangent length from C is $CE = s - c = 3k$, $IE \perp CA$, and CI bisects angle C, so $\angle ICE = \dfrac{C}{2}$.

In the right triangle CEI: $CI = \dfrac{CE}{\cos\frac{C}{2}} = \dfrac{3k\sqrt{14}}{3} = \sqrt{14}\,k$. (Equivalently $r = (s-c)\tan\dfrac{C}{2} = \sqrt{5}\,k$ and $CI = \dfrac{r}{\sin\frac{C}{2}}$.)

Given $CI = 2\sqrt{14}$: $k = 2$, so the sides are $14, 16, 18$.

Perimeter $= 2s = 24k = 48$.

Final Answer: 48

Question 06

English

In a triangle ABC, let a, b, c denote the lengths of the sides opposite to the angles A, B, C, respectively. Suppose $c\cos B = 1$, $b\cos C = 3$ and $\sin B = 2\sin C$. Then the value of $\cos A$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक त्रिभुज ABC में माना a, b, c क्रमशः कोणों A, B, C की सम्मुख भुजाओं की लम्बाइयाँ हैं। माना $c\cos B = 1$, $b\cos C = 3$ तथा $\sin B = 2\sin C$ है। तब $\cos A$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Projection formula on side a: $a = b\cos C + c\cos B = 3 + 1 = 4$.

Sine law: $\dfrac{b}{c} = \dfrac{\sin B}{\sin C} = 2$, so $b = 2c$.

Projection formulae on sides b and c: $b = a\cos C + c\cos A$ and $c = a\cos B + b\cos A$.

Multiply the first by b and the second by c: $b^2 = a\,b\cos C + bc\cos A$ and $c^2 = a\,c\cos B + bc\cos A$.

Subtracting: $b^2 - c^2 = a(b\cos C - c\cos B) = 4(3 - 1) = 8$.

With $b = 2c$: $3c^2 = 8$, so $c^2 = \dfrac{8}{3}$, $b^2 = \dfrac{32}{3}$ and $bc = 2c^2 = \dfrac{16}{3}$.

From $b^2 = ab\cos C + bc\cos A$: $\dfrac{32}{3} = 4 \cdot 3 + \dfrac{16}{3}\cos A$, so $\dfrac{16}{3}\cos A = -\dfrac{4}{3}$ and $\cos A = -\dfrac{1}{4}$.

Check: $a = 4$, $b = \dfrac{4\sqrt{6}}{3} \approx 3.27$, $c = \dfrac{2\sqrt{6}}{3} \approx 1.63$ form a triangle, and $\cos B = \dfrac{1}{c} > 0$, $\cos C = \dfrac{3}{b} < 1$ are consistent; the cosine rule also gives $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc} = \dfrac{40/3 - 16}{32/3} = -\dfrac{1}{4}$.

So $\cos A = -0.25$.

Final Answer: -0.25

Question 07

English

In a triangle ABC, let a, b, c denote the lengths of the sides opposite to the angles A, B, C, respectively. The incircle of the triangle has radius 2 and touches the side AB at the point F with $AF = 3$. If the area of the triangle ABC is 24, then the value of $a^2 - b^2 - c^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक त्रिभुज ABC में माना a, b, c क्रमशः कोणों A, B, C की सम्मुख भुजाओं की लम्बाइयाँ हैं। त्रिभुज के अन्तःवृत्त की त्रिज्या 2 है तथा यह भुजा AB को बिन्दु F पर स्पर्श करता है, जहाँ $AF = 3$ है। यदि त्रिभुज ABC का क्षेत्रफल 24 है, तो $a^2 - b^2 - c^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The tangent length from A to the incircle is $AF = s - a$, so $s - a = 3$.

Half-angle formula: $\tan\dfrac{A}{2} = \sqrt{\dfrac{(s-b)(s-c)}{s(s-a)}} = \dfrac{\sqrt{s(s-a)(s-b)(s-c)}}{s(s-a)} = \dfrac{\Delta}{s(s-a)} = \dfrac{r}{s-a} = \dfrac{2}{3}$.

Sub-multiple angle formulae with $t = \tan\dfrac{A}{2} = \dfrac{2}{3}$: $\cos A = \dfrac{1-t^2}{1+t^2} = \dfrac{5/9}{13/9} = \dfrac{5}{13}$ and $\sin A = \dfrac{2t}{1+t^2} = \dfrac{4/3}{13/9} = \dfrac{12}{13}$.

Area: $\dfrac{1}{2}bc\sin A = 24$, so $bc = \dfrac{48 \cdot 13}{12} = 52$.

Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A$, so $a^2 - b^2 - c^2 = -2bc\cos A = -2 \cdot 52 \cdot \dfrac{5}{13} = -40$.

Existence: $s = \dfrac{\Delta}{r} = 12$, $a = s - 3 = 9$, $b + c = 15$, $bc = 52$; b, c are the roots of $x^2 - 15x + 52 = 0$, which are real and positive ($225 - 208 = 17 > 0$), and $a = 9 < b + c$.

So $a^2 - b^2 - c^2 = -40$ (indeed $81 - (225 - 104) = -40$).

Final Answer: -40

Question 08

English

Consider triangles ABC in which the sides opposite to the angles A and B have lengths $a = 15$ and $b = 20$, respectively, and $\cos A = \frac{4}{5}$. There are exactly two non-congruent such triangles; let γ_1 and γ_2 denote the measures of the angle C in these two triangles. Then the value of $\sin\gamma_1 + \sin\gamma_2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

ऐसे त्रिभुजों ABC पर विचार कीजिए जिनमें कोणों A तथा B की सम्मुख भुजाओं की लम्बाइयाँ क्रमशः $a = 15$ तथा $b = 20$ हैं, और $\cos A = \frac{4}{5}$ है। ऐसे ठीक दो असर्वांगसम त्रिभुज हैं; माना γ_1 तथा γ_2 इन दोनों त्रिभुजों में कोण C के माप हैं। तब $\sin\gamma_1 + \sin\gamma_2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A is acute with $\cos A = \frac{4}{5}$, so $\sin A = \frac{3}{5}$.

Sine law: $\sin B = \frac{b\sin A}{a} = \frac{20 \cdot \frac{3}{5}}{15} = \frac{4}{5}$.

Ambiguous case: $b\sin A = 12$ and $12 < a = 15 < b = 20$, so there are two triangles. Let B_1 be the acute solution ($\cos B_1 = \frac{3}{5}$) and $B_2 = \pi - B_1$.

Both are admissible: $B_1 + A < \pi$ clearly, and $B_2 + A < \pi \iff A < B_1 \iff \cos A > \cos B_1$, which holds since $\frac{4}{5} > \frac{3}{5}$.

Then $\gamma_1 = \pi - A - B_1$, so $\sin\gamma_1 = \sin(A + B_1) = \frac{3}{5}\cdot\frac{3}{5} + \frac{4}{5}\cdot\frac{4}{5} = 1$ (a right angle at C).

And $\gamma_2 = \pi - A - (\pi - B_1) = B_1 - A$, so $\sin\gamma_2 = \sin(B_1 - A) = \frac{4}{5}\cdot\frac{4}{5} - \frac{3}{5}\cdot\frac{3}{5} = \frac{7}{25}$.

Hence $\sin\gamma_1 + \sin\gamma_2 = 1 + \frac{7}{25} = \frac{32}{25} = 1.28$ (equivalently $2\sin B_1\cos A$).

Check with the sides: $c^2 - 2bc\cos A + b^2 - a^2 = 0$ gives $c = 25$ or $c = 7$, and $\sin C = \frac{c\sin A}{a}$ gives 1 and $\frac{7}{25}$.

Final Answer: 1.28

Question 09

English

Let $B = (0,0)$, $C = (7,0)$ and $A = (2,4)$. The circle having the line segment BC as a diameter meets the sides AB and AC again at the points F and E, respectively, where $F \neq B$ and $E \neq C$. Then the circumradius of the triangle AEF is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $B = (0,0)$, $C = (7,0)$ तथा $A = (2,4)$ हैं। रेखाखण्ड BC को व्यास मानकर खींचा गया वृत्त भुजाओं AB तथा AC को पुनः क्रमशः बिन्दुओं F तथा E पर काटता है, जहाँ $F \neq B$ तथा $E \neq C$ है। तब त्रिभुज AEF की परित्रिज्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The circle with BC as diameter is $(x-0)(x-7)+(y-0)(y-0)=0$, i.e. $x^2+y^2-7x=0$, centre $\left(\frac{7}{2},0\right)$, radius $\frac{7}{2}$.

F: line AB is $y=2x$; substituting, $5x^2-7x=0$, so $x=\frac{7}{5}$ and $F=\left(\frac{7}{5},\frac{14}{5}\right)$; it lies between A and B.

E: line AC is $4x+5y=28$; with $y=\frac{28-4x}{5}$ the circle gives $41x^2-399x+784=0=(x-7)(41x-112)$, so $E=\left(\frac{112}{41},\frac{140}{41}\right)$, between A and C.

Since E, F lie on the circle with diameter BC, $\angle BEC=\angle BFC=\frac{\pi}{2}$: $BE \perp AC$ and $CF \perp AB$ are altitudes; let them meet at the orthocentre H.

Then $\angle AEH=\angle AFH=\frac{\pi}{2}$, so E and F lie on the circle with AH as diameter, which is therefore the circumcircle of triangle AEF; its radius is $\frac{AH}{2}$.

Orthocentre: the altitude from A is $x=2$ (as BC is the x-axis); the altitude from B is perpendicular to AC (direction $(5,-4)$), i.e. $5x-4y=0$. At $x=2$: $y=\frac{5}{2}$, so $H=\left(2,\frac{5}{2}\right)$.

$AH=4-\frac{5}{2}=\frac{3}{2}$, so the circumradius of triangle AEF is $\frac{3}{4}$.

Check: the circle with diameter AH is $(x-2)^2+(y-4)\left(y-\frac{5}{2}\right)=0$; for F: $\left(-\frac{3}{5}\right)^2+\left(-\frac{6}{5}\right)\left(\frac{3}{10}\right)=\frac{9}{25}-\frac{9}{25}=0$. Also $R\cos A=\frac{\sqrt{20}\cdot\sqrt{41}\cdot 7}{4\cdot 14}\cdot\frac{6}{\sqrt{20}\sqrt{41}}=\frac{3}{4}$.

So the circumradius is 0.75.

Final Answer: 0.75

Question 10

English

In a triangle ABC, let a, b, c denote the lengths of the sides opposite to the angles A, B, C, respectively. The incircle of the triangle and the excircle opposite to the vertex A touch the side BC at the points P and Q, respectively. If the inradius is 7, the radius of the excircle opposite to A is 12 and $PQ = 8$, then the area of the triangle ABC is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक त्रिभुज ABC में माना a, b, c क्रमशः कोणों A, B, C की सम्मुख भुजाओं की लम्बाइयाँ हैं। त्रिभुज का अन्तःवृत्त तथा शीर्ष A के सम्मुख बाह्यवृत्त भुजा BC को क्रमशः बिन्दुओं P तथा Q पर स्पर्श करते हैं। यदि अन्तःत्रिज्या 7 है, शीर्ष A के सम्मुख बाह्यवृत्त की त्रिज्या 12 है तथा $PQ = 8$ है, तो त्रिभुज ABC का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $x = s - b$ and $y = s - c$. The incircle touches BC at P with $BP = s - b = x$; the excircle opposite to A touches BC at Q with $BQ = s - c = y$.

Hence $PQ = |x - y| = |c - b| = 8$, and $a = BC = x + y$.

Inradius $r = \dfrac{\Delta}{s}$ and exradius $r_1 = \dfrac{\Delta}{s-a}$, so $r\, r_1 = \dfrac{\Delta^2}{s(s-a)} = \dfrac{s(s-a)(s-b)(s-c)}{s(s-a)} = xy$.

So $xy = 7 \cdot 12 = 84$ and $|x - y| = 8$; then $(x+y)^2 = (x-y)^2 + 4xy = 64 + 336 = 400$, so $x + y = 20$ and $\{x, y\} = \{6, 14\}$.

Thus $a = 20$.

Also $\dfrac{r_1}{r} = \dfrac{s}{s-a} = \dfrac{12}{7}$, so $7s = 12(s - 20)$, giving $s = 48$.

Area $\Delta = rs = 7 \cdot 48 = 336$.

Check: $b = s - x$, $c = s - y$ are $42, 34$ (in some order); Heron: $\sqrt{48 \cdot 28 \cdot 6 \cdot 14} = \sqrt{112896} = 336$, and $\dfrac{336}{28} = 12 = r_1$.

Final Answer: 336

Question 11

English

Let ABC be a triangle with the vertex $A = (-6, 4)$, circumcentre $O = (0, 1)$ and orthocentre $H = (-3, 1)$. Let D be the foot of the perpendicular from A to the side BC. If $D = (\alpha, \beta)$, then the value of $\alpha + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना ABC एक त्रिभुज है जिसका शीर्ष $A = (-6, 4)$, परिकेन्द्र $O = (0, 1)$ तथा लम्बकेन्द्र $H = (-3, 1)$ है। माना D, A से भुजा BC पर डाले गए लम्ब का पाद है। यदि $D = (\alpha, \beta)$ है, तो $\alpha + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On the Euler line, the centroid G divides OH internally in the ratio $1 : 2$, so $G = \dfrac{2O + H}{3} = \left(\dfrac{0-3}{3}, \dfrac{2+1}{3}\right) = (-1, 1)$.

G divides the median from A in the ratio $2 : 1$, so the midpoint M of BC satisfies $G = \dfrac{A + 2M}{3}$: $M = \dfrac{3G - A}{2} = \left(\dfrac{-3+6}{2}, \dfrac{3-4}{2}\right) = \left(\dfrac{3}{2}, -\dfrac{1}{2}\right)$.

The altitude from A passes through H, so $AH \perp BC$. $\overrightarrow{AH} = (3, -3)$ has slope -1, hence BC has slope 1; through M: $y + \dfrac{1}{2} = x - \dfrac{3}{2}$, i.e. $y = x - 2$.

(Consistent with $OM \perp BC$: $\overrightarrow{OM} = \left(\dfrac{3}{2}, -\dfrac{3}{2}\right)$.)

The foot D lies on line AH: $y - 4 = -(x + 6)$, i.e. $y = -x - 2$.

Intersection with $y = x - 2$: $x = 0$, $y = -2$, so $D = (0, -2)$, a vertex of the pedal triangle.

Check: B, C are the points of $y = x - 2$ at distance $OA = \sqrt{45}$ from O: $(6, 4)$ and $(-3, -5)$; then $A + B + C - 2O = (-3, 1) = H$, and the triangle is acute, so D lies on the side BC.

Hence $\alpha + \beta = 0 + (-2) = -2$.

Final Answer: -2

Question 12

English

In a triangle ABC, $BC = 8$ and the length of the median AD is $\sqrt{34}$. The internal and the external bisectors of the angle BAC meet the line BC at the points P and Q, respectively, and $PQ = 6$. Then the area of the triangle ABC is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक त्रिभुज ABC में $BC = 8$ है तथा माध्यिका AD की लम्बाई $\sqrt{34}$ है। कोण BAC के आन्तरिक तथा बाह्य समद्विभाजक रेखा BC को क्रमशः बिन्दुओं P तथा Q पर मिलते हैं, तथा $PQ = 6$ है। तब त्रिभुज ABC का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $k = \dfrac{AB}{AC}$. The external bisector meets line BC only if $AB \neq AC$, so $k \neq 1$.

By the bisector theorems, $\dfrac{PB}{PC} = \dfrac{QB}{QC} = k$: P divides BC internally and Q externally in the ratio $k : 1$, so P, Q are harmonic conjugates with respect to B, C.

Place BC on a number line with $B = 0$, $C = 8$. Section formula: $P = \dfrac{k \cdot 8 + 1 \cdot 0}{k+1} = \dfrac{8k}{k+1}$, $Q = \dfrac{k \cdot 8 - 1 \cdot 0}{k-1} = \dfrac{8k}{k-1}$.

Distance: $PQ = \left| \dfrac{8k}{k-1} - \dfrac{8k}{k+1} \right| = \dfrac{16k}{|k^2-1|}$.

For $k < 1$: $\dfrac{16k}{1-k^2} = 6 \Rightarrow 3k^2 + 8k - 3 = 0 \Rightarrow (3k-1)(k+3) = 0 \Rightarrow k = \dfrac{1}{3}$. For $k > 1$ the same equation gives $k = 3$; the two cases are mirror images (swap B and C).

Take $AC = 3AB$. Length of median (Apollonius): $AB^2 + AC^2 = 2AD^2 + 2\left(\dfrac{BC}{2}\right)^2 = 68 + 32 = 100$.

So $10AB^2 = 100$: $AB = \sqrt{10}$, $AC = 3\sqrt{10}$ (a valid triangle since $\sqrt{10} + 8 > 3\sqrt{10}$).

Cosine rule at B: $\cos B = \dfrac{10 + 64 - 90}{16\sqrt{10}} = -\dfrac{1}{\sqrt{10}}$, so $\sin B = \dfrac{3}{\sqrt{10}}$.

Area $= \dfrac{1}{2} \cdot AB \cdot BC \cdot \sin B = \dfrac{1}{2} \cdot \sqrt{10} \cdot 8 \cdot \dfrac{3}{\sqrt{10}} = 12$.

Check with coordinates: $B = (0,0)$, $C = (8,0)$, $A = (-1,3)$ gives $AB = \sqrt{10}$, $AC = \sqrt{90}$, $AD^2 = 25 + 9 = 34$, $P = (2,0)$, $Q = (-4,0)$, $PQ = 6$, area $= \dfrac{1}{2} \cdot 8 \cdot 3 = 12$.

Final Answer: 12

Question 13

English

Let $A = (2, 1)$ and $B = (5, 5)$, and let C be a point on the line $y = 1$ such that the angle BAC is obtuse. The internal bisector of the angle BAC meets the side BC at the point D. If $AD = \dfrac{2\sqrt{5}}{3}$, then the x-coordinate of C is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A = (2, 1)$ तथा $B = (5, 5)$ हैं, तथा C रेखा $y = 1$ पर एक ऐसा बिन्दु है कि कोण BAC अधिक कोण है। कोण BAC का आन्तरिक समद्विभाजक भुजा BC को बिन्दु D पर मिलता है। यदि $AD = \dfrac{2\sqrt{5}}{3}$ है, तो C का x-निर्देशांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Line AB: slope $\dfrac{4}{3}$ through A, i.e. $4x - 3y - 5 = 0$, and $c = AB = \sqrt{9 + 16} = 5$. Line AC is $y = 1$; write $C = (2 + t, 1)$.

$\cos \angle BAC = \dfrac{(3, 4) \cdot (t, 0)}{5|t|} = \dfrac{3t}{5|t|}$, which is negative only for $t < 0$. So $C = (2 - b, 1)$ with $b = AC > 0$, and $\cos A = -\dfrac{3}{5}$.

The bisectors of the angles between $4x - 3y - 5 = 0$ and $y - 1 = 0$ are $\dfrac{4x - 3y - 5}{5} = \pm(y - 1)$, i.e. $x - 2y = 0$ and $2x + y - 5 = 0$.

The bisector of $\angle BAC$ has direction $\dfrac{(3, 4)}{5} + (-1, 0) = \left(-\dfrac{2}{5}, \dfrac{4}{5}\right) \propto (-1, 2)$, so it is $2x + y - 5 = 0$ (the other one bisects the acute angle).

Half-angle: the angle between $(-1, 2)$ and $(-1, 0)$ has $\cos \dfrac{A}{2} = \dfrac{1}{\sqrt{5}}$ (consistent with $\cos A = 2 \cdot \dfrac{1}{5} - 1 = -\dfrac{3}{5}$).

Length of the angle bisector: $AD = \dfrac{2bc}{b + c} \cos \dfrac{A}{2} = \dfrac{2 \cdot 5b}{b + 5} \cdot \dfrac{1}{\sqrt{5}}$.

Setting $\dfrac{10b}{\sqrt{5}(b + 5)} = \dfrac{2\sqrt{5}}{3}$: $30b = 10(b + 5)$, so $b = \dfrac{5}{2}$ (unique, as $\dfrac{10b}{b + 5}$ is increasing).

Hence $C = \left(2 - \dfrac{5}{2}, 1\right) = \left(-\dfrac{1}{2}, 1\right)$.

Check: D divides BC in $AB : AC = 2 : 1$, $D = \dfrac{1 \cdot B + 2 \cdot C}{3} = \left(\dfrac{4}{3}, \dfrac{7}{3}\right)$, which lies on $2x + y - 5 = 0$, and $AD^2 = \dfrac{4}{9} + \dfrac{16}{9} = \dfrac{20}{9}$.

So the x-coordinate of C is -0.5.

Final Answer: -0.5

Question 14

English

Let ABC be a triangle with $AB = AC = 5$ and $BC = 6$, and let I_1, I_2, I_3 be its excentres opposite to the vertices A, B, C, respectively. Let Δ_1 denote the area of the triangle ABC and Δ_2 the area of the triangle $I_1I_2I_3$. Then the value of $\dfrac{\Delta_1}{\Delta_2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना ABC एक त्रिभुज है जिसमें $AB = AC = 5$ तथा $BC = 6$ है, तथा I_1, I_2, I_3 क्रमशः शीर्षों A, B, C के सम्मुख इसके बाह्यकेन्द्र हैं। माना Δ_1 त्रिभुज ABC का क्षेत्रफल तथा Δ_2 त्रिभुज $I_1I_2I_3$ का क्षेत्रफल है। तब $\dfrac{\Delta_1}{\Delta_2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The excentres I_2 and I_3 lie on the external bisector of angle A, and I_1 lies on the internal bisector AI. Internal and external bisectors are perpendicular, so $I_1A \perp I_2I_3$ with A on I_2I_3. Similarly $I_2B \perp I_3I_1$ and $I_3C \perp I_1I_2$.

Hence A, B, C are the feet of the altitudes of triangle $I_1I_2I_3$: triangle ABC is its pedal triangle, and the incentre I (common point of I_1A, I_2B, I_3C) is its orthocentre.

Angles of the excentral triangle: $\angle I_2I_1I_3 = \dfrac{\pi}{2} - \dfrac{A}{2}$, and similarly for the others; all are acute.

For an acute triangle with angles α, β, γ and circumradius R', the pedal triangle has sides $R' \sin 2\alpha$, $R' \sin 2\beta$, $R' \sin 2\gamma$ and angles $\pi - 2\alpha$, $\pi - 2\beta$, $\pi - 2\gamma$, so its area is $\dfrac{1}{2}R'^2 \sin 2\alpha \sin 2\beta \sin 2\gamma$; the triangle's area is $2R'^2 \sin\alpha \sin\beta \sin\gamma$. The ratio is $2\cos\alpha\cos\beta\cos\gamma$.

With $\cos\left(\dfrac{\pi}{2} - \dfrac{A}{2}\right) = \sin\dfrac{A}{2}$ etc.: $\dfrac{\Delta_1}{\Delta_2} = 2\sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2} = 2 \cdot \dfrac{r}{4R} = \dfrac{r}{2R}$.

For $AB = AC = 5$, $BC = 6$: the altitude from A is 4, $\Delta = 12$, $s = 8$, $r = \dfrac{\Delta}{s} = \dfrac{3}{2}$, $R = \dfrac{abc}{4\Delta} = \dfrac{150}{48} = \dfrac{25}{8}$.

So $\dfrac{\Delta_1}{\Delta_2} = \dfrac{3/2}{25/4} = \dfrac{6}{25} = 0.24$.

Check: with $A = (0, 4)$, $B = (-3, 0)$, $C = (3, 0)$ the excentres are $I_1 = (0, -6)$, $I_2 = (5, 4)$, $I_3 = (-5, 4)$, so $\Delta_2 = \dfrac{1}{2} \cdot 10 \cdot 10 = 50$ and $\dfrac{\Delta_1}{\Delta_2} = \dfrac{12}{50} = 0.24$.

Final Answer: 0.24